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Appendix

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A Limitations

While this study provides some insightful findings in the field of in-context learning, there are some limitations that should be noted. First, the experiments in this work constrain themselves to repeating surface patterns. More sophisticated patterns are not discussed. Second, our work mainly focuses on revealing the token co-occurrence reinforcement and understanding its influence on in-context learning. Its detrimental influences on in-context learning suggest that resolving the spurious connections would be helpful to either chain-of-thought or in-context learning.

B Experimental Details

B.1 Dataset Descriptions

In this paper, we mainly use the following five datasets, and we introduce each of them and describe our preprocess of these datasets individually. We present cases from each dataset to demonstrate their characteristics in Table 2.

Wikitext-103 The Wikitext-103 dataset, introduced by Merity et al. (2016)[4], is a language modeling dataset that contains a collection of over 100 million tokens extracted from the set of verified Good and Featured articles on Wikipedia. We use a randomly sampled collection of 1000 sentences provided by Xu et al. (2022)[5]. The provided version is pre-tokenized to words and we use the moses detokenizer[6] to restore the untokenized version for compatibility with the tokenizer of transformers.

[4] `https://blog.salesforceairesearch.com/the-wikitext-long-term-dependency-language-modeling-`
[5] `https://github.com/Jxu-Thu/DITTO/blob/main/data/wiki_sentences.txt`
[6] `https://github.com/moses-smt/mosesdecoder/blob/master/scripts/tokenizer/detokenizer.perl`

Table 2: Datasets and Cases

Dataset	**Number of Cases**	**Examples**
Wikitext-103	1000	_The Bintulu Government Secondary School was built in 1964._
BookCorpus	1000	_“A little, ”he admitted._
Random	1000	_Green Ou incarcer hijab ura na Edmonton regardless iken Mayor_
GSM8K	1000	_Question: Janet’s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers’ market daily for $2 per fresh duck egg. How much in dollars does she make every day at the farmers’ market?_ _Let’s think step by step._ _Janet sells 16 - 3 - 4 = «16-3-4=9»9 duck eggs a day. She makes 9 * 2 = $«9*2=18»18 every day at the farmer’s market. ### 18_
MMLU	1140	_As more lamps are connected in parallel in a circuit, the current in the power source_ _A. increases_ _B. decreases_ _C. remains the same_ _D. Not enough information to say_ _Answer: A_